

# J-Anne Yow

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## Summary

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Final-year PhD candidate in Robotics at NTU Singapore, developing intelligent robotic systems spanning perception, planning and control. Experienced in integrating large language models and reinforcement learning for adaptive manipulation. My goal is to build scalable, human-aware robot intelligence for deployment in real-world environments.

## Education

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**Nanyang Technological University**, Singapore

Sept 2021 – Feb 2026

Ph.D. in Mechanical Engineering

(Expected)

- **Supervisor:** Prof Ang Wei Tech
- **Dissertation:** Towards Personalized Robot Assistance: Integrating User Preferences in Robot-Assisted Feeding
- **Research Interests:** Human-Robot Interaction, Machine Learning, Foundation Models, Continual Learning

**Nanyang Technological University**, Singapore

Aug 2017 – May 2021

BE in Mechanical Engineering with a Second Major in Business

- **GPA:** 4.92/5.00, Dean's List every Academic Year
- **Awards:** Dr Leung Shiu Kee Gold Medal Award, ASEAN Undergraduate Scholarship

## Research Experience

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**Robot-Assisted Feeding**

2022 – present

- Led the development of an end-to-end robot-assisted feeding system, integrating perception, planning, and control for safe, adaptive bite sequencing, food acquisition and bite transfer.
- Developed a language-driven robot adaptation pipeline, leveraging large language models to map user feedback into constrained motion parameters to ensure safe and predictable robot behavior.
- Developed a goal-conditioned reinforcement learning policy for adaptive manipulation across variable food properties, enabling the acquisition of targeted bite sizes.
- Deployed the system in elderly user trials, analyzed user feedback to guide design improvements and refine robot behavior for safe and effective assistance.

**Grasping in Clutter**

2020 – 2023

- Developed a point-and-click interface for human-in-the-loop robotic manipulation, integrating object segmentation, grasp ranking, and grasp cycling to streamline user input and improve selection efficiency in cluttered scenes.
- Developed a POMDP-based shared autonomy framework that queries users under intent uncertainty, reducing conflicting actions and enabling faster task completion.

## Work Experience

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**Research Assistant**, Nanyang Technological University – Singapore

Aug 2021 – present

- Led the technical delivery of a multi-year robotics project, overseeing integration of perception, planning and control modules into a safe, deployable robot-assisted feeding system for real-world user trials.
- Managed project planning, resource coordination, and ethical compliance, removing operational bottlenecks such as compute limitations to keep development on schedule.
- Mentored and scoped eight undergraduate projects, defining research objectives and guiding skill development.
- Drafted and contributed to competitive funding proposals to support ongoing research and new initiatives.

## Selected Publications

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**SAVR: Scooping Adaptation for Variable food properties via Reinforcement Learning**

Oct 2025

*J-Anne Yow*, Wei Tech Ang

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

**FRANC: Feeding Robot for Adaptive Needs and Personalized Care** Oct 2025  
*J-Anne Yow*, Luke Toh, Yi Heng San, Wei Tech Ang  
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

**ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework** Oct 2025  
Sherwin Chan, *J-Anne Yow*, Yi Heng San, Vasanth Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang  
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

**Shared Autonomy of a Robotic Manipulator for Grasping under Human Intent Uncertainty using POMDPs** Nov 2023  
*J-Anne Yow*, Neha P Garg, Wei Tech Ang  
IEEE Transactions on Robotics (T-RO)

## Service

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**Reviewer:** IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA)

## Skills

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**Technical Skills:** Python, C/C++, PyTorch, Git, SQL

**Robotics Skills:** ROS/ROS2, MoveIt, MuJoCo, Isaac Lab, Motion Planning, Computer Vision, Reinforcement Learning, Human-Robot Interaction

**Languages:** English, Chinese, Malay